

OS LAB MANUAL (CS23431)

Roll No:230701229

EX.NO:8

PRODUCER CONSUMER USING SEMAPHORES

Aim: To write a C program to do Inter Process Communication (IPC) using shared memory between sender process and receiver process

Program:

```
#include<stdio.h>

#include<stdlib.h>

int mutex = 1, full = 0, empty = 3, x = 0;

int wait(int s) {

    return (--s);

}

int signal(int s) {

    return (++s);

}

void producer() {

    mutex = wait(mutex);

    full = signal(full);

    empty = wait(empty);

    x++;

    printf("\nProducer produces item %d", x);

    mutex = signal(mutex);

}

void consumer() {

    mutex = wait(mutex);

    full = wait(full);

    empty = signal(empty);

    printf("\nConsumer consumes item %d", x);

    x--;
```

```
mutex = signal(mutex);
}

int main() {

    int n;

    printf("\n1.Producer\n2.Consumer\n3.Exit");
    while(1) {

        printf("\nEnter your choice: ");
        scanf("%d", &n);
        switch(n) {

            case 1:

                if ((mutex == 1) && (empty != 0))
                    producer();

                else
                    printf("Buffer is full!!");

                break;

            case 2:

                if ((mutex == 1) && (full != 0))
                    consumer();

                else
                    printf("Buffer is empty!!");

                break;

            case 3:

                exit(0);

            default:

                printf("Invalid choice!");

        }

    }

    return 0;
}
```

Input:

```
1.Producer
2.Consumer
3.Exit
```

Output:

```
1.Producer
2.Consumer
3.Exit
Enter your choice:1

Producer produces the item 1
Enter your choice:2

Consumer consumes item 1
Enter your choice:1

Producer produces the item 1
Enter your choice:1

Producer produces the item 2
Enter your choice:2

Consumer consumes item 2
Enter your choice:2

Consumer consumes item 1
Enter your choice:2
Buffer is empty!!
```